## 1. Alice is trying to access a website using her web browser.

She enters the website's domain name in the address bar. Explain the process of how her computer resolves the domain name to an IP address and accesses the website.

ANS: When Alice enters a website's domain name (e.g. www.example.com), her computer uses the Domain Name System (DNS) to convert it into an IP address (like 192.0.2.1) which identifies the web server.

1. The browser first checks its local cache for the IP.
2. If not found, it queries the DNS resolver (usually provided by the ISP).
3. The resolver contacts the root DNS server, which directs it to the Top-Level Domain (TLD) server (.com, .org, etc.).
4. The TLD server points to the Authoritative Name Server for example.com, which returns the IP address.
5. The browser then uses this IP to send an HTTP/HTTPS request to the web server.
6. The web server responds with the requested webpage, which is rendered on Alice's browser.

## 2. What are the parts of a Domain Name? Give one example and describe the parts.

ANS: A domain name typically has three parts:

1. Subdomain – Identifies a specific section (e.g. www or mail).
2. Second-Level Domain (SLD) – The unique website name (e.g. google).
3. Top-Level Domain (TLD) – The category or type of domain (e.g. .com).

Example: www.google.com

- www → Subdomain
- google → Second-Level Domain
- .com → Top-Level Domain

## 3. Name three domain names of websites that fall under:

3.1 Government (.gov)

ANS: india.gov.in, nasa.gov, usa.gov

3.2 Organization (.org)

ANS: wikipedia.org, unicef.org, redcross.org

3.3 Education (.edu)

ANS: mit.edu, stanford.edu, harvard.edu

3.4 Commercial (.com)

ANS: amazon.com, google.com, flipkart.com

3.5 Network Providers / Computers (.net)

ANS: speedtest.net, slideshare.net, sourceforge.net

3.6 International Databases or Treaties (.int)

ANS: un.int, nato.int, eu.int

## 4. Bob is setting up a new web server for his online business.

He needs to choose between HTTP and HTTPS for secure communication. Explain the differences between HTTP and HTTPS and recommend which one Bob should use.

ANS:

- HTTP (HyperText Transfer Protocol) transfers data in plain text, which can be intercepted by hackers.
- HTTPS (HyperText Transfer Protocol Secure) uses SSL/TLS encryption, ensuring secure data transmission between browser and server.

Recommendation:

Bob should use HTTPS as it protects user data, builds customer trust, improves SEO ranking, and is essential for secure online transactions.

## 5. Mark is working on a website with multiple pages.

He wants to create links between these pages. Explain what a relative URL is, provide an example, and describe when Mark should use relative URLs instead of absolute ones.

ANS: A Relative URL gives the path to a file relative to the current page's location, without specifying the full domain name.

Example: If the current page is www.example.com/about/index.html, and you want to link to the contact page in the same folder, use:

<a href="contact.html">Contact Us</a>

Use Case:

Relative URLs are ideal when linking within the same website because they make it easier to move the website to a new domain or server without breaking links.